HOSPITAL QUEUE INTELLIGENCE

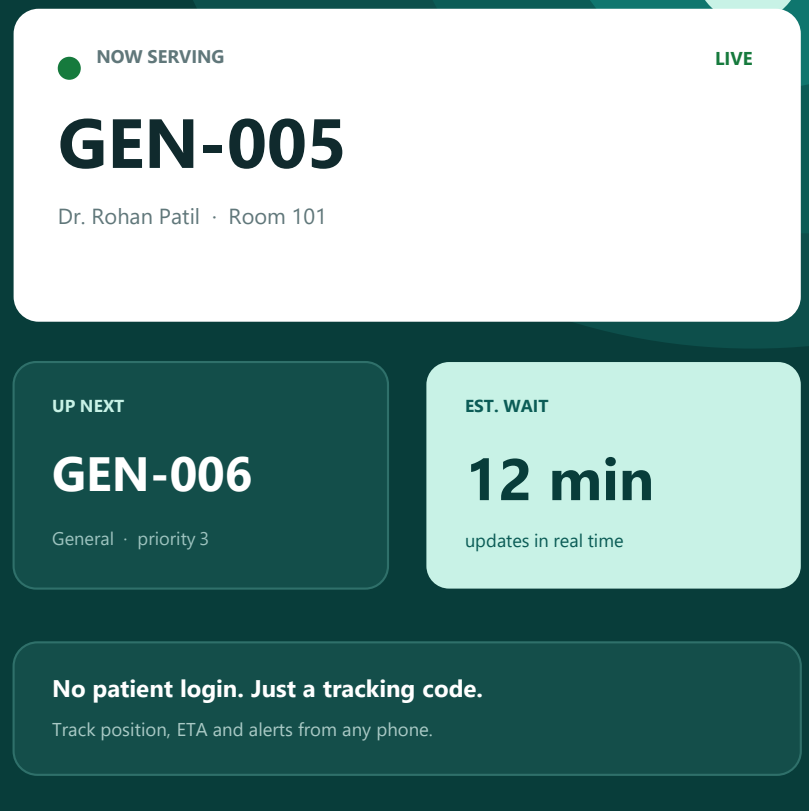
Queue less. Care more.

SQMS turns hospital waiting rooms into a live, fair, multilingual queue experience - from booking to consultation.

SMART QUEUE TIME MANAGEMENT SYSTEM

Project presentation / 2026

REACT EXPRESS SOCKET.IO SQLITE



THE CHALLENGE

Waiting is not the problem. Uncertainty is.

Hospitals already manage queues. The gap is information, fairness and coordination.

01

Waiting blind

Patients do not know their position, expected wait or whether the doctor is running late.

02

Manual priority

Emergency and vulnerable patients are often prioritised verbally, without a consistent audit trail.

03

Fragmented roles

Reception, nursing and doctors work from separate lists, paper tokens or ad-hoc messaging.

04

No visibility

Managers can see activity, but not trends, bottlenecks, no-shows or service performance.

Patients do not need to be first. They need to know when - and staff need one reliable queue to act on.

THE SOLUTION

One connected queue, from token to consultation.

SQMS combines patient self-service, live queue intelligence, role-based controls and operational analytics.

THE SHIFT

From a room full of uncertainty to a single source of truth.

- No patient account required
- Priority is visible and consistent
- Every role sees the same live state
- Every action is measurable

1

Book without login

Same-day walk-in tokens or appointments up to 30 days ahead.

2

Follow live position

Token status, people ahead, predicted wait and estimated consultation time.

3

Run the clinic

Doctor controls, front-desk check-in, triage, pause and recall workflows.

4

Reach every patient

English, Hindi and Marathi alerts through SMS, in-app and voice.

PATIENT JOURNEY

A clear path with no account and no guesswork.

A patient can take a token in under a minute, then track it from any browser.

1

Book or walk in

Choose department, doctor, priority and language.

2

Get a token

Receive a code such as GEN-261029-006.

3

Track live

See position, people ahead and a rolling ETA.

4

Check in

Self check-in for today's appointment, or arrive for a walk-in.

5

Consult

Called, started and completed through the doctor's queue.

EXAMPLE TOKEN

GEN-006

Position 3 · 12 min wait · Room 101

WORKS ON ANY PHONE

The tracking link opens directly - no app install, no patient login.

QUEUE INTELLIGENCE

Priority that is fair. Waiting time that is explainable.

The queue is re-ordered whenever the underlying state changes - not just when a screen refreshes.

SERVICE ORDER

1 **Emergency**
Immediate clinical need
FIRST

2 **Senior · pregnant · differently abled**
Vulnerable-patient tier
SECOND

3 **General**
Routine and follow-up visits
THIRD

Tie-breaker: first come, first served within the same tier.

LIVE ETA MODEL

Position and wait recalculated in real time.

ETA = remaining time
+ patients ahead x doctor average

- New check-in
- Priority change
- Skip or recall
- Pause or resume

The same model powers patient tracking and the public display.

ROLE-BASED WORKFLOWS

Every role gets exactly the controls it needs.

Permissions are enforced in the interface, the API and the audit trail.

ADMIN

Full oversight

- Analytics & exports
- All queues & controls
- Staff/access management

SCOPE

Hospital-wide

DOCTOR

Own clinic flow

- Call next or specific
- Start / complete / skip
- Pause, resume, recall

SCOPE

Own patients

NURSE

Front desk + triage

- Check-in & booking
- Priority updates
- Patient alerts

SCOPE

Department

RECEPTION

Registration desk

- Book appointments
- Check in & cancel
- Search and alert

SCOPE

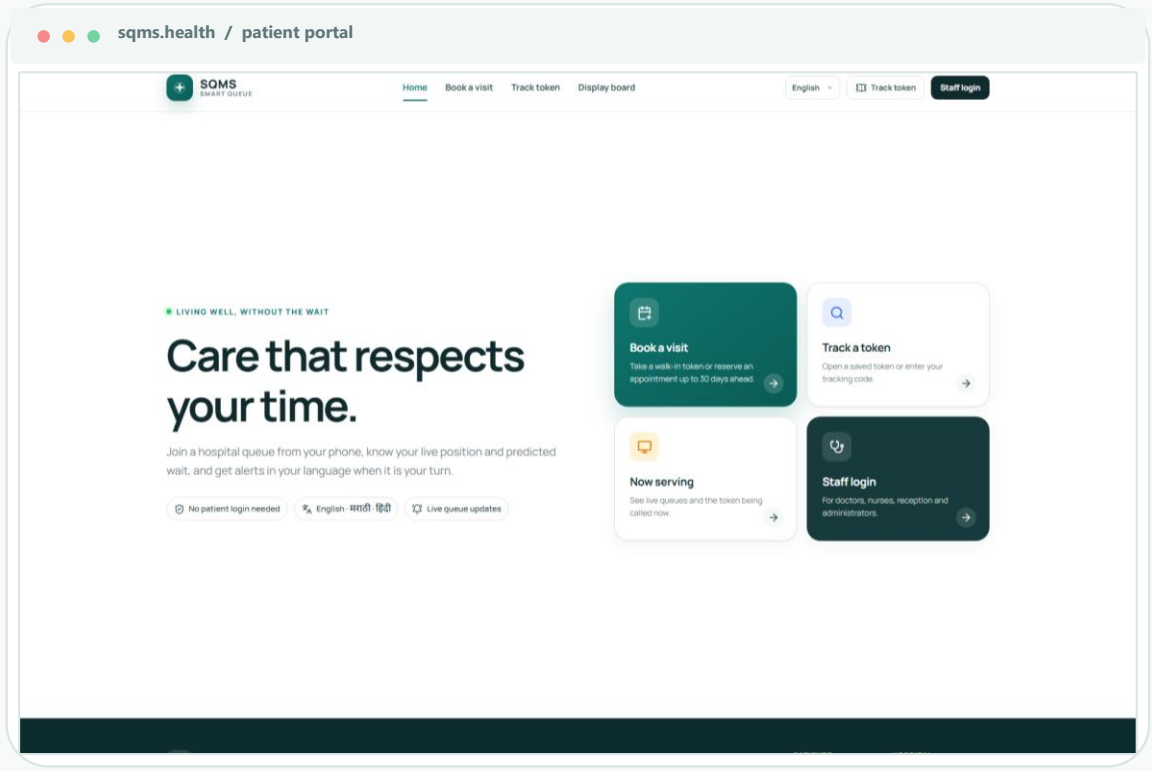
Department

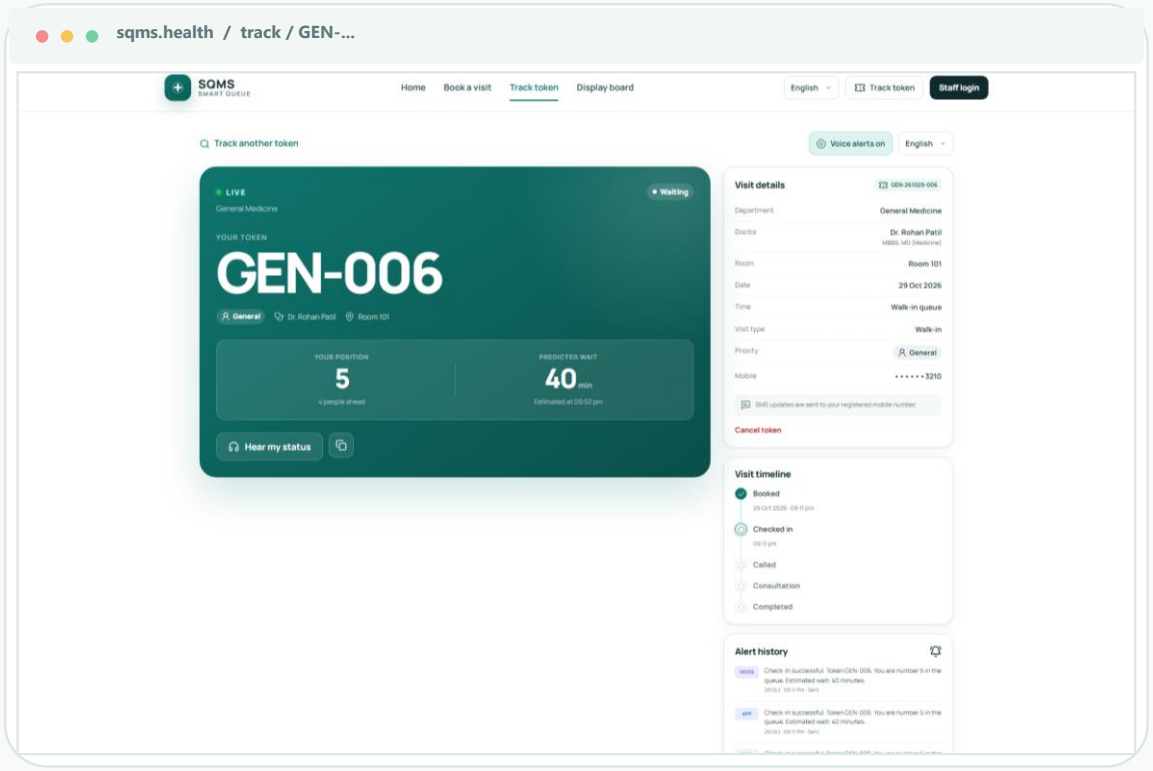
One live queue state · consistent permissions · auditable actions

PATIENT EXPERIENCE

Book in seconds. Track without an account.

A responsive public portal makes the queue visible before the patient enters the hospital.





Booking without friction

Department, doctor, date, priority and language in one guided flow.

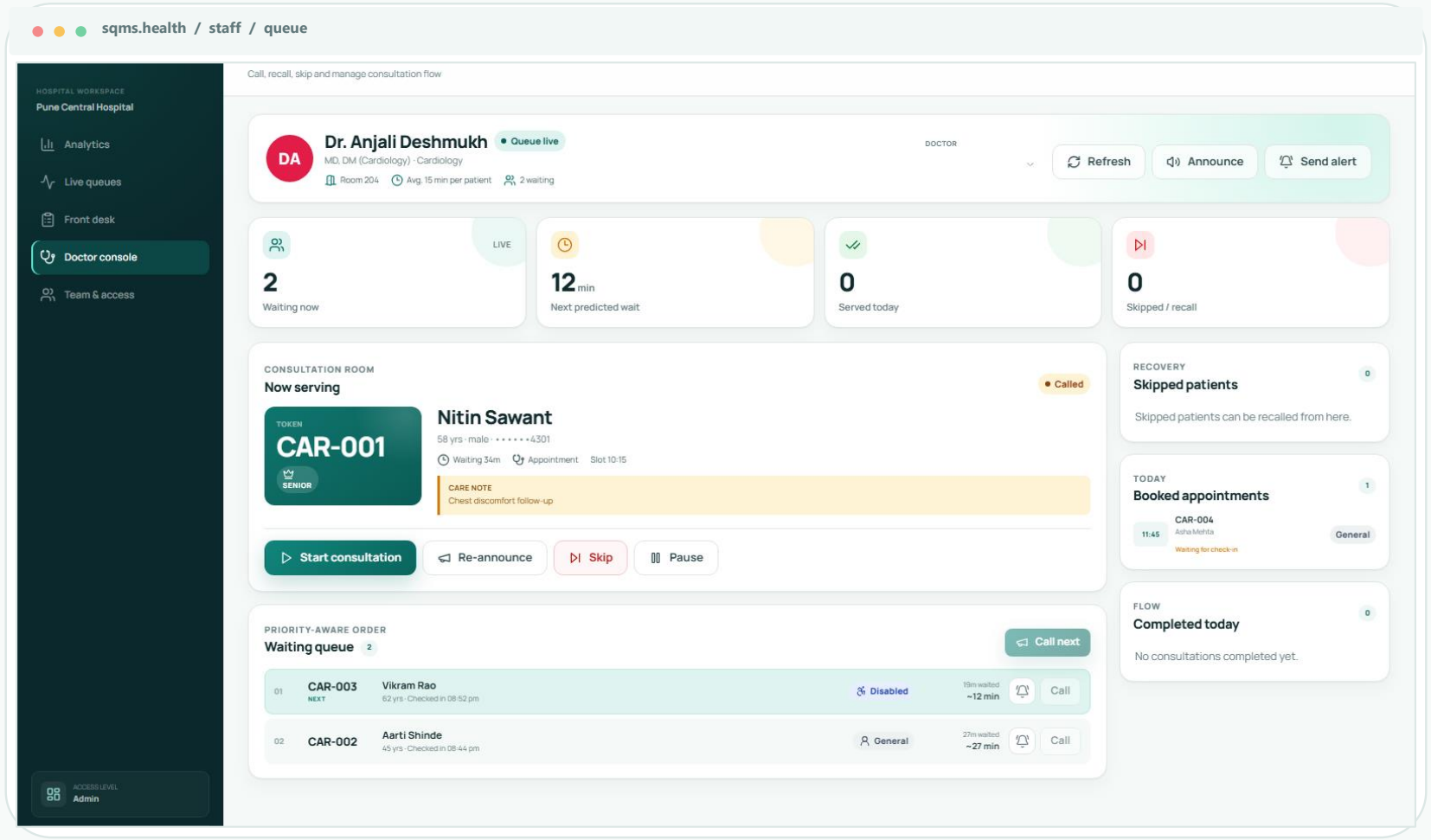
Confidence while waiting

Live token, people ahead, ETA, room, timeline and notification history.

DOCTOR WORKSPACE

The entire clinic flow on one screen.

Current patient, waiting queue, booked patients and served history update live.



DOCTOR CONTROLS

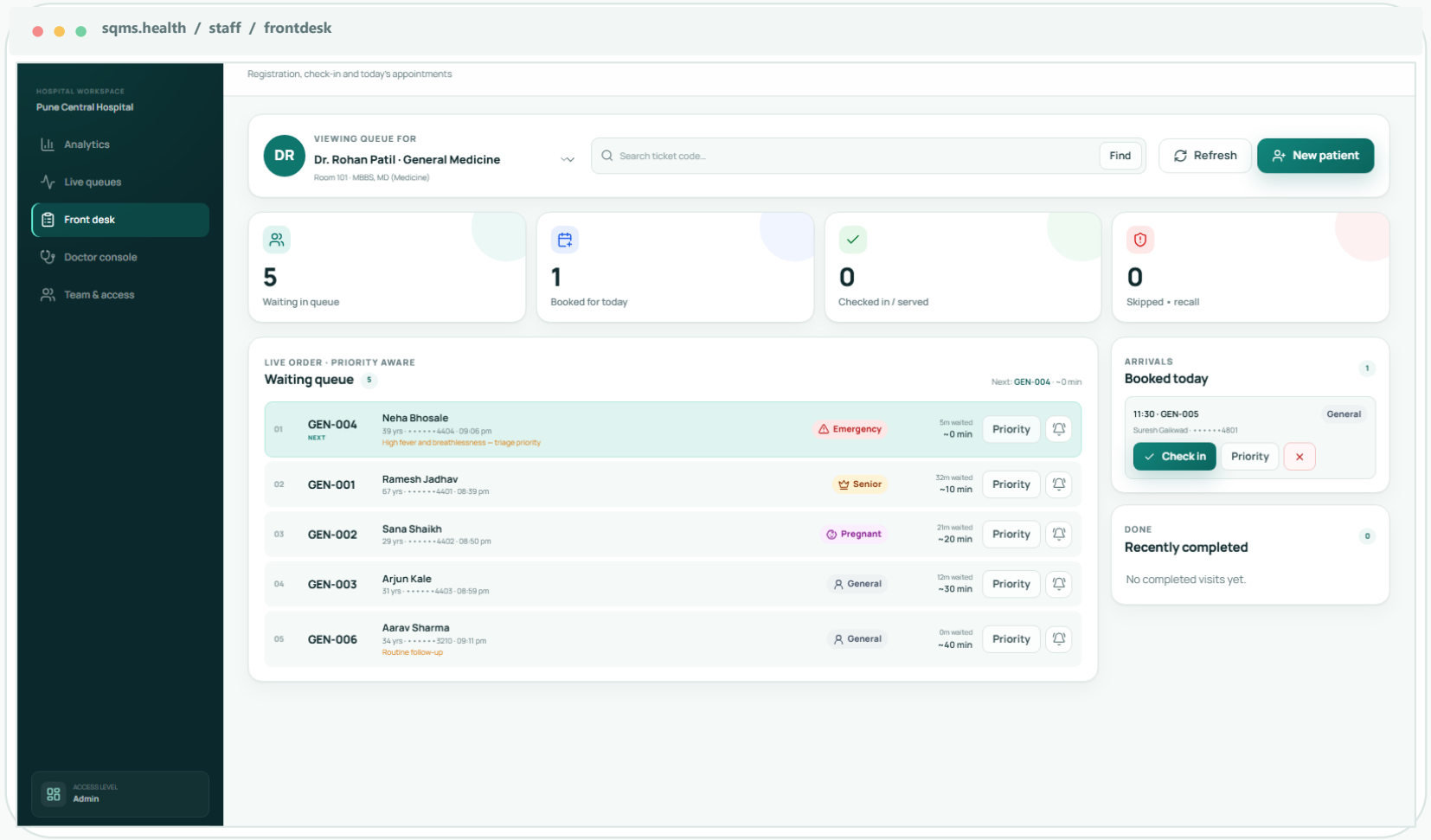
Act on the queue without leaving the patient context.

- **Call next**
  Auto-selects the highest-priority waiting patient.
- **Call specific**
  Bring a particular token forward.
- **Start / complete**
  Maintain accurate consultation timing.
- **Skip / recall**
  Handle no-shows without losing the ticket.
- **Pause / resume**
  Communicate delays to everyone waiting.

FRONT DESK

Registration, check-in and triage without lost tickets.

Reception and nursing share one operational view, with permissions that match their responsibilities.



ONE SEARCH

Find any patient by ticket code.

Check in, update priority, send an alert or cancel - from the same result.

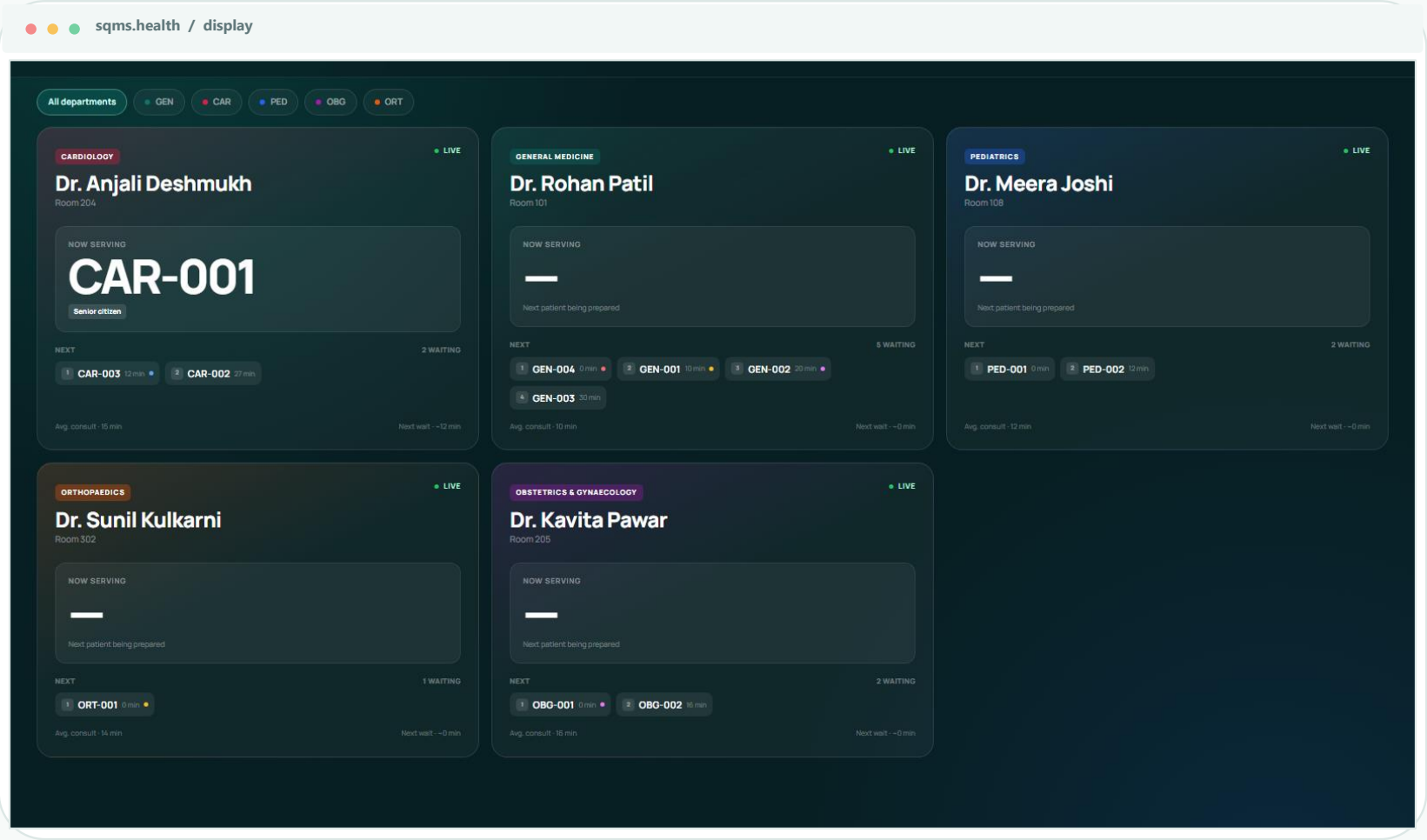
DESK CONTROLS

- Create walk-in or appointment tickets
- Self check-in support for patients
- Triage priority with audit history
- Cancel bookings and send notices

PUBLIC DISPLAY

A waiting room that can see what is happening.

A full-screen, multilingual board keeps patients informed and reduces repeated desk queries.



MULTILINGUAL BY DESIGN

One queue. Three languages.

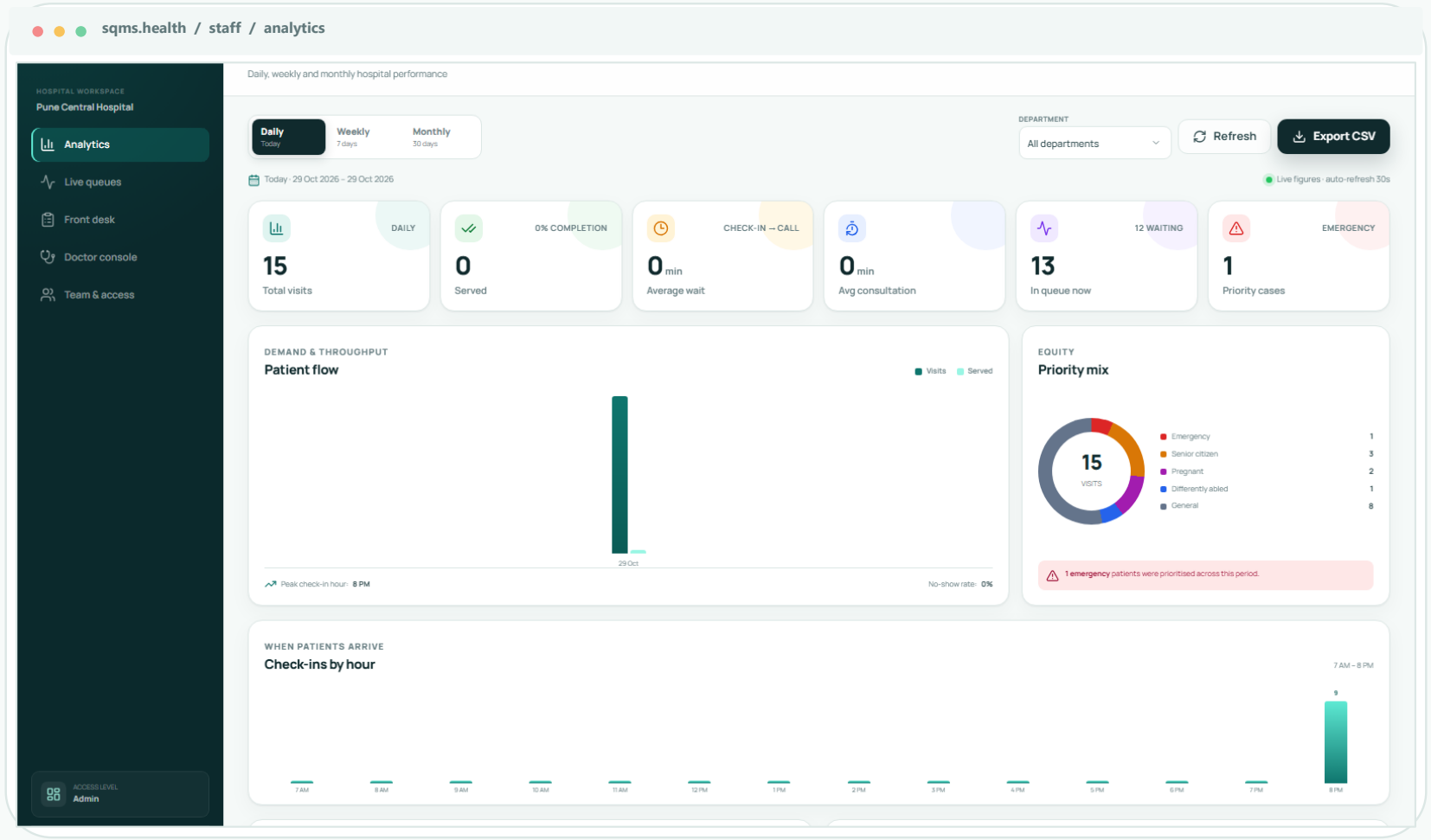
ENGLISH	English
MARATHI	मराठी
HINDI	हिन्दी

Auto-refresh + live socket updates keep the board current.

ANALYTICS

Turn daily operations into decisions.

Daily, weekly and monthly views expose demand, service performance and bottlenecks.



Daily

Today's flow, peak hours and priority mix

Weekly

Seven-day demand and service trend

Monthly

Department and doctor performance

ALSO INCLUDED

Department filters · CSV export
Audit log · notification outbox

COMMUNICATION

Alerts that reach people in their language.

Templates are generated from the same live queue state and delivered through multiple channels.

ENGLISH

1 **Turn alert**

Token GEN-006, please proceed to Room 101.

MARATHI

1 **Turn alert**

टोकन GEN-006, कृपया खोली 101 मध्ये या.

HINDI

1 **Turn alert**

टोकन GEN-006, कृपया कमरा 101 में आएं.

DELIVERY CHANNELS

SMS

Twilio when configured; simulated and logged otherwise.

IN-APP

Persistent notification history inside the tracking page.

VOICE

Browser speech synthesis for waiting-room and device alerts.

Templates: booked · checked in · your turn · delay · skipped · resumed · cancelled

ARCHITECTURE

A compact stack built for live state.

React handles the experience. Express owns the rules. Socket.IO keeps every screen in sync.



LIVE SYNC queue:changed · board:changed · alert:voice

DELIVERY SMS via Twilio · In-app outbox · Speech synthesis

TRUST bcrypt passwords · Role guards · Audit log

TRUST & QUALITY

Small stack, serious safeguards.

The demo is not just visually complete: authentication, authorization, data integrity and runtime flows were validated.

JWT SESSIONS

Signed tokens with role and doctor scope; expired sessions are rejected.

ROLE ENFORCEMENT

The same guards protect routes, queue actions and analytics endpoints.

AUDIT TRAIL

Booking, priority, call, skip, pause, alert and staff actions are recorded.

DATA INTEGRITY

Foreign keys, WAL mode, transaction-safe state changes and validations.

VALIDATED DELIVERY

Ready to demo.

29/29	21/21	0
API checks	Browser checks	Audit vulnerabilities

Build passes, production server serves the SPA, and public + staff flows were exercised in headless Chrome.

Deployment note: replace demo credentials, set a strong JWT secret, use HTTPS and configure Twilio for real SMS delivery.

RUN THE DEMO

One command to start. Four roles to explore.

The database resets to a clean, seeded demo with live queues and 45 days of history.

TERMINAL

```
$ npm install
$ npm run dev

Web   http://localhost:5173
API   http://localhost:4000
```

PRODUCTION

```
$ npm run build && npm start
```

Reset demo data
npm run reset-db --workspace=@sqms/api

DEMO ACCOUNTS

Admin	admin@sqms.health	admin123	**Doctor**	cardio@sqms.health	doctor123
Nurse	nurse@sqms.health	nurse123	**Reception**	reception@sqms.health	reception123

PUBLIC URLS

Patient portal	/
Token tracking	/track/:code
Display board	/display
Staff login	/staff/login

SOURCE

github.com/202501110031-cmd/sqms

WHAT'S NEXT

A platform, not just a queue.

NOW	Live queue, booking, alerts, analytics and role controls
NEXT	Postgres/cloud deployment, reminders, kiosk mode and multi-hospital tenancy
LATER	ML-based wait prediction, EHR integration and patient feedback loops

SQMS IN ONE LINE

A live, fair and multilingual queue that gives patients certainty and gives staff control.

Patient portal · staff workflows
analytics · alerts

Better queues. Better care.

Thank you / Questions